

Discrete Structure

BCA — Semester 2 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Give any four examples that are not propositions. Assume the premises, all oversmart persons are stupid, children of stupid persons are naughty, John is oversmart, Sam is children of John. Using rules of inferences, show that Sam is naughty.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Why do we need principle of inclusion and exclusion? How many ways can we express the four character words ending with a digit and beginning three are lowercase alphabets. Be sure that none of the character can be repeated. Using induction, show that $3^n - 1$ is multiple of 2 for $n \geq 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. State necessary and sufficient conditions for a graph to have Euler path and circuit. Find the GCD of 12 and 18 using Extended Euclidian Algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Show that the sum of two even numbers is even using direct proof.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define power set. What is the power set of the set $A = \{1, 2, 3, 4\}$?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain one-to-one correspondence with example. What is identity function?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Use mathematical induction to prove that the sum of the first n odd positive integers is n^2 .

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is recursively defined function? Suppose that f is defined recursively by $f(0) = 3$, $f(n + 1) = 2f(n) + 3$. Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is arithmetic modulo m ? Use the definition of addition and multiplication in $\mathbb{Z}_m$ to find $7 + 9$ and $7 * 9$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define equivalence relation with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is generalized pigeonhole principle. If a class has 24 students, what is the maximum number of possible grading that must be done to ensure that there at least two students with the same grade.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write the Dijkstra's algorithm to find the shortest path between two nodes in graph.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.